Unit 7, Lesson 2: Slope in Real-World Contexts

Benchmarks

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

MA.8.AR.3.5 Given a real-world context, determine and interpret the slope and *y*-intercept of a two-variable linear equation from a written description, a table, a graph, or an equation in slope-intercept form.

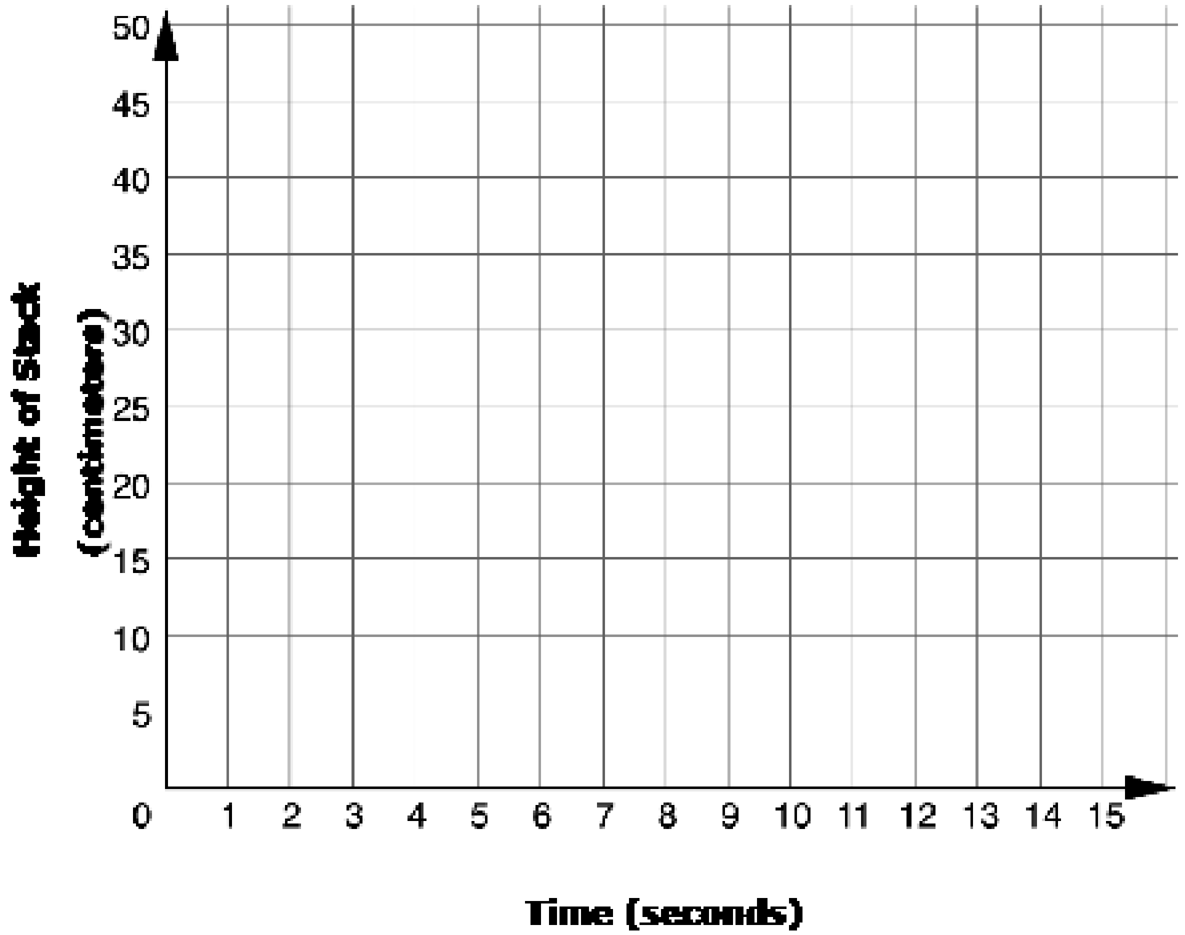
Learning Targets

- ☐ I can determine the slope from a graph, table, or written description in a real-world context.
- ☐ I can interpret the slope as the rate of change in a real-world context.

7.2.1 Warm-Up: Graph the Story

Watch the video of the person stacking styrofoam cups.

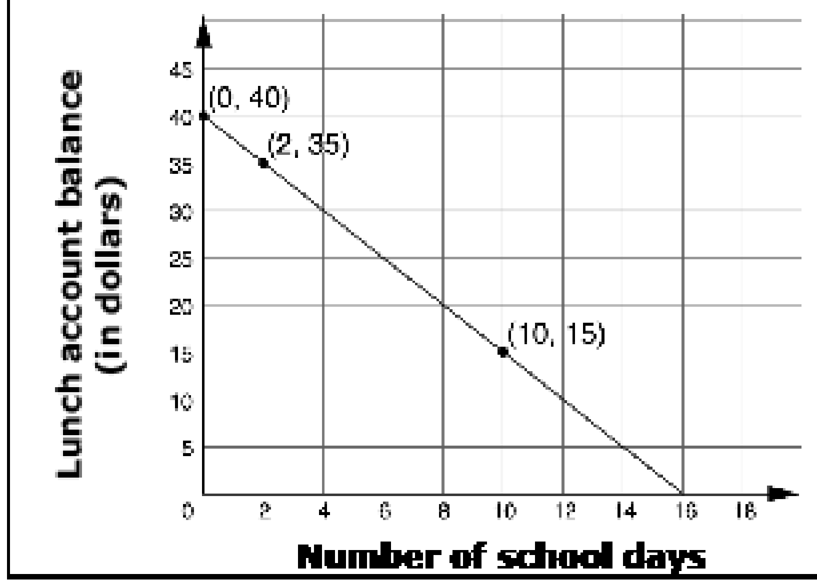
1. Create a sketch of the graph to represent the height of the stack, in centimeters, over time.



7.2.2 Guided Instruction: Finding and Interpreting Slope in Context

A linear relationship is any relationship between two quantities, where one quantity has a constant rate of change with respect to the other. This means that one quantity is increasing or decreasing the same amount for every equal-sized interval of the other quantity.

Linear relationships can be used to model real-world contexts in which the quantities are changing at a constant rate. Three different representations of the same linear relationship are shown.

<table>
<tr><th colspan="2">Written Description</th></tr>
<tr><td colspan="2">Lavar has a $40 balance available on his lunch account. Each school day, he spends $2.50 from his account to buy lunch.</td></tr>
<tr><th>Graph</th><th>Table of Values</th></tr>
<tr><td>
</td>
<td><table>
<tr><th>x</th><th>y</th></tr>
<tr><td>0</td><td>40</td></tr>
<tr><td>1</td><td>37.5</td></tr>
<tr><td>2</td><td>35</td></tr>
<tr><td>3</td><td>32.5</td></tr>
<tr><td>4</td><td>30</td></tr>
</table></td></tr>
</table>

1. Use the graph to find the slope of the line.

2. Use the table to find the slope of the line.

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The written description of the relationship can also be used to find the slope.

3. Use the written description to complete the statements describing how to find the slope.

The balance of Lavar's lunch account is [o increasing / o decreasing] each school day, so the slope is [o positive. / o negative.] It changes by [o $40 / o $2.50] each school day, so the slope is [o 2.5. / o −2.5. / o 40. / o −40.]

Tabitha earns money mowing lawns each weekend. She charges customers $15 per hour to mow their lawn, in addition to the $7 flat fee she charges them for gasoline and other supplies she'll need.

4. Two values are given in the context. Explain which value represents the rate of change.

5. Determine whether the rate of change represents an increase or decrease in the amount of money Tabitha earns. Show your work or explain your answer.

6. Based on the answers to the previous questions, what is the slope in this linear relationship?

7. Complete the statement to interpret the slope as the rate of change for this context.

In this context, the slope means that the amount of money Tabitha earns [o increases / o decreases] by [o 7 / o 15] [o dollars / o hours] for every [o dollar / o hour] she mows a customer's lawn.

The **slope** is the **rate of change** that one quantity is changing by in terms of another quantity. When interpreting the rate of change in a real-world context, **units for both quantities** are included.

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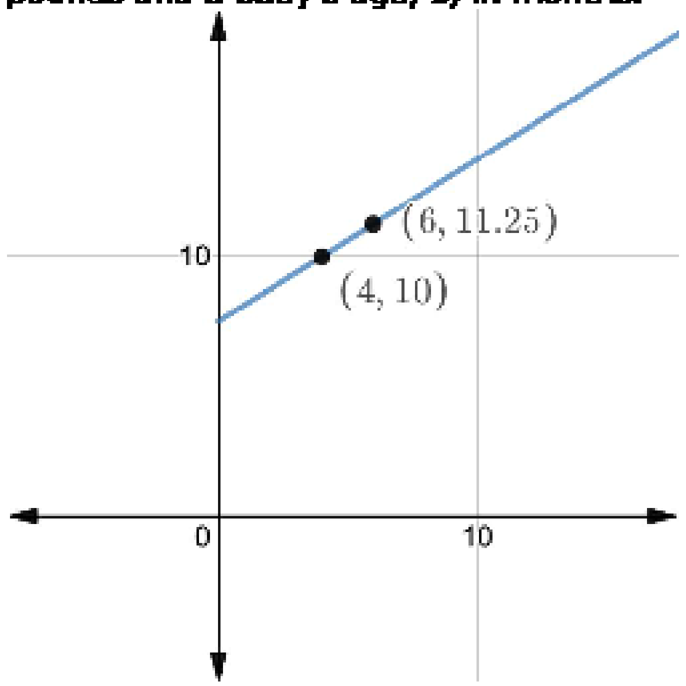
7.2.3 Guided Practice: Relating Slope to Rate of Change in Real-World Context

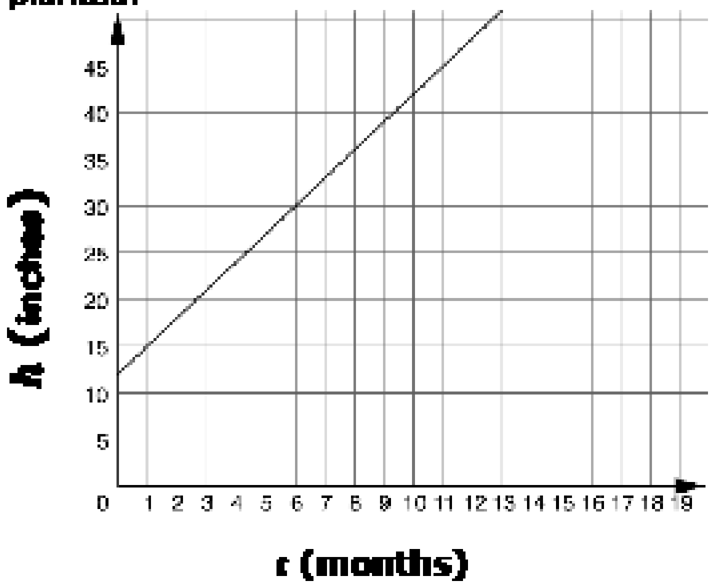
For each linear relationship, determine the rate of change and interpret its meaning in the given context.

<table>
<tr><th>Linear Relationship</th><th>Rate of Change</th><th>Interpretation</th></tr>
<tr><td>The table of values represents the relationship between the cost, c, in dollars of s, bags of sour patch candies.
<table>
<tr><th>s</th><th>c</th></tr>
<tr><td>2</td><td>2.5</td></tr>
<tr><td>3</td><td>3.75</td></tr>
<tr><td>4</td><td>5</td></tr>
</table></td><td></td><td></td></tr>
<tr><td>The graph represents the relationship between the average baby's weight, y, in pounds and a baby's age, x, in months.

</td><td></td><td></td></tr>
</table>

Linear Relationship	Rate of Change	Interpretation
A mountain road gains elevation at a constant rate. After 2 miles, the elevation is 5,500 feet above sea level. After 4 miles, the elevation is 6,200 feet above sea level.		

7.2.4 Your Turn!

For each linear relationship, determine the rate of change and interpret its meaning in the given context.

Linear Relationship	Rate of Change	Interpretation
The graph shows the height in inches, h, of a bamboo plant t months after it has been planted. 		
Lin is reading a book for an assignment in English. After reading the first 30 pages, she makes a plan to read 40 pages each day until she finishes the book.		

7.2.5 Wrap-Up: Ticket Out the Door

Yingjie tracked the cost of purchasing gasoline and the number of gallons he bought. His table is shown where x represents the number of gallons purchased, and y represents the cost in dollars.

x	y
5.4	19.71
6	21.90
15	54.75
17	62.05

1. What is the rate of change?

2. Interpret the rate of change.